

AQUA-SEEKER

Active Groundwater Exploration Agent

Deciding **what to test**
before **where to drill**.

Soundous Talha · Noor Abuznadah · Tala Alzaeem · Sadeen Alshammari

CS3081-1 **Artificial Intelligence**

Instructor: **Dr. Naila Marir**

October 2026



THE PROBLEM

Groundwater exploration is expensive because every drilling decision is made **under uncertainty**.

70%

of Saudi Arabia is classified as arid

Source: MEWA

80%

of freshwater use is for agriculture

Source: MEWA

THE CONSEQUENCE

A dry well wastes drilling investment and delays access to usable water.

OUR AIM

Build an agent that reaches a usable well with fewer tests and **fewer dry wells**.



EXISTING SOLUTIONS

Current approaches produce ranked locations, but the decision of **what to test next** is missing.

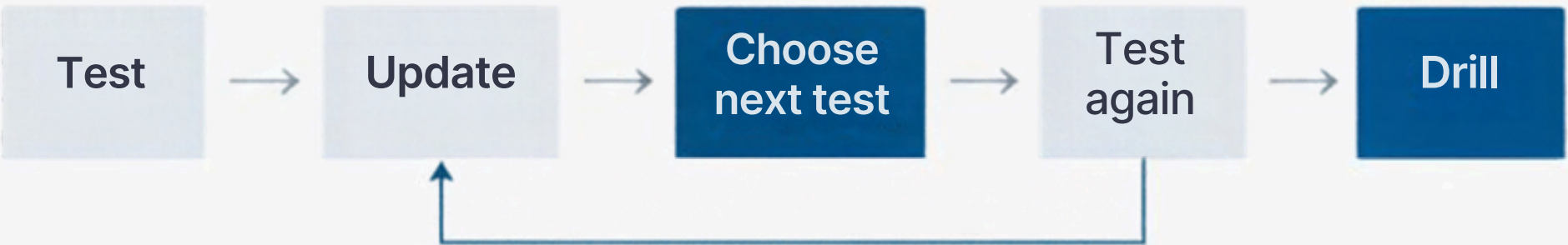
POTENTIAL MAPPING	Combines satellite, geological, and terrain data to rank promising zones.
GEOPHYSICAL SURVEYS	Provides stronger underground evidence with higher cost per site.
EXPLORATORY BOREHOLES	Delivers reliable evidence but is the most expensive option.



CURRENT APPROACH



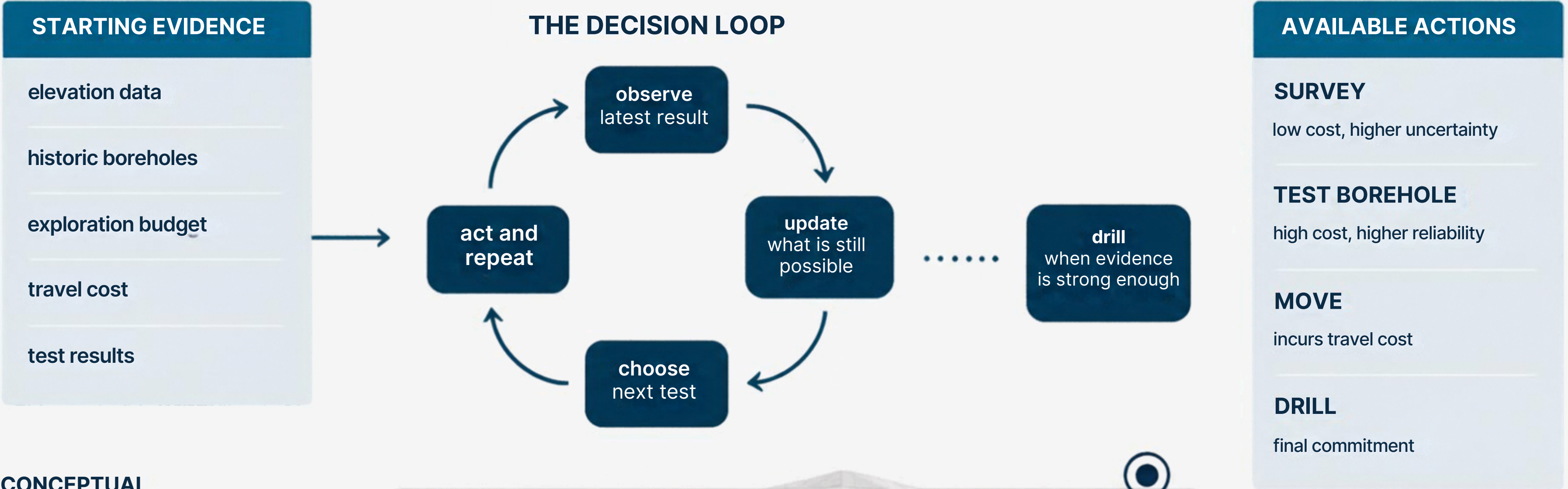
AQUA-SEEKER APPROACH



Existing systems identify promising locations.
AQUA-SEEKER decides **what evidence should be collected next**.

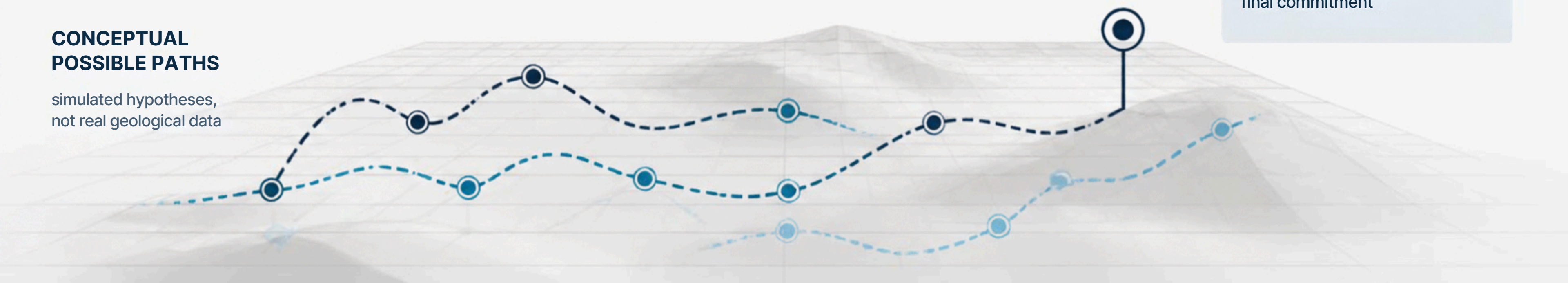
AQUA-SEEKER

active agent for groundwater exploration



CONCEPTUAL POSSIBLE PATHS

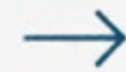
simulated hypotheses, not real geological data



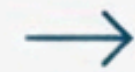
WORKED EXAMPLE

The agent plans tests, learns from each result, and stops when more testing would cost more than it would reveal.

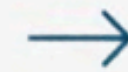
01 Initial uncertainty
20 possible hidden groundwater channels



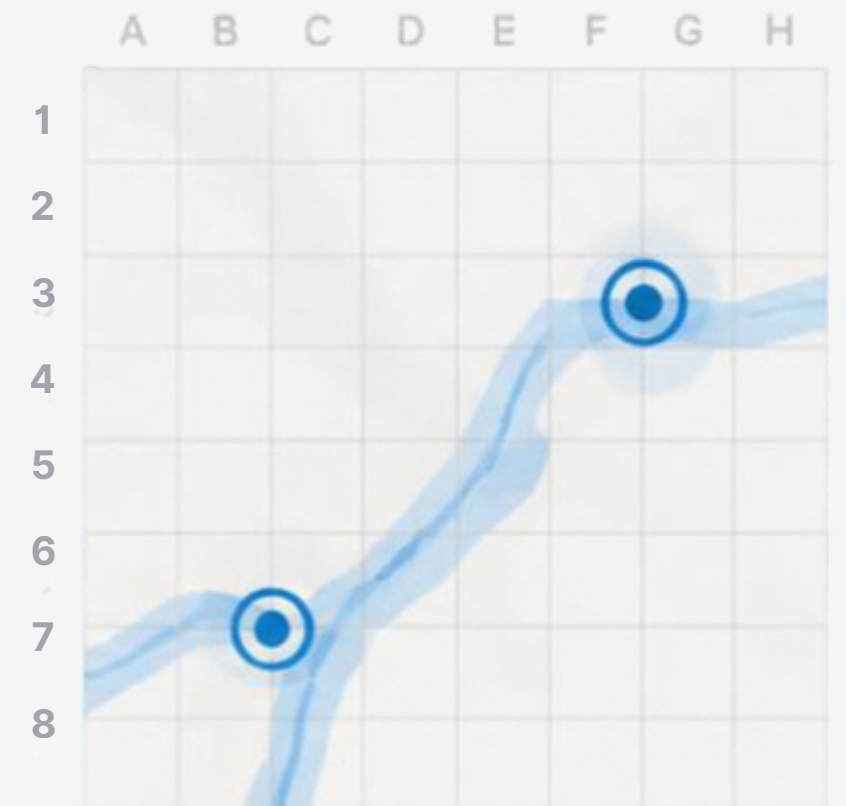
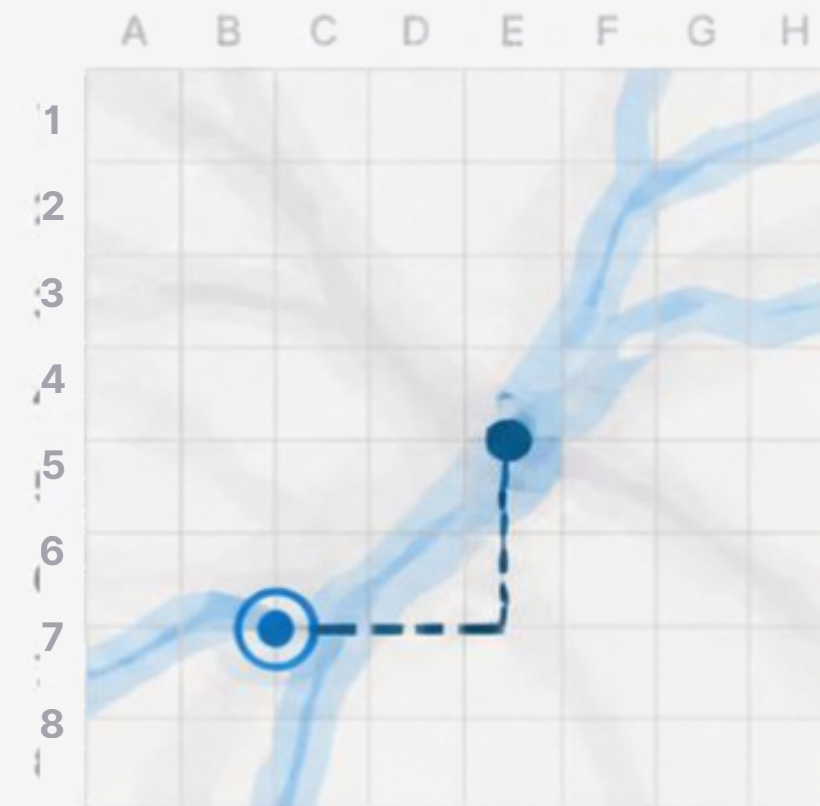
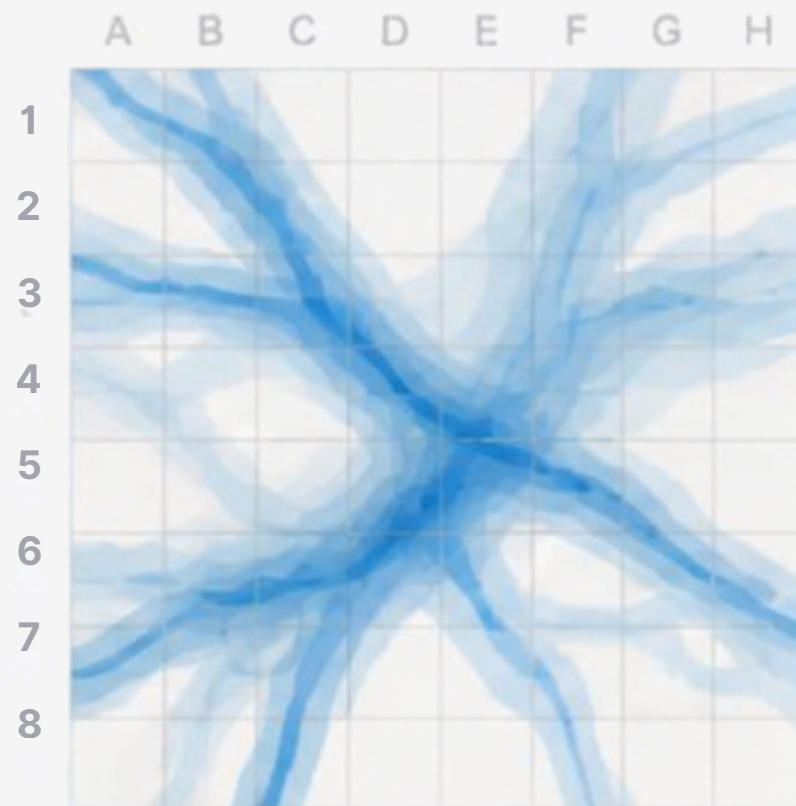
02 Survey at C7
Wet signal. Updates likelihood, clay may also read wet.



03 Borehole at E6
Dry result removes every hypothesis through E6.
A* routes to lowest cost.

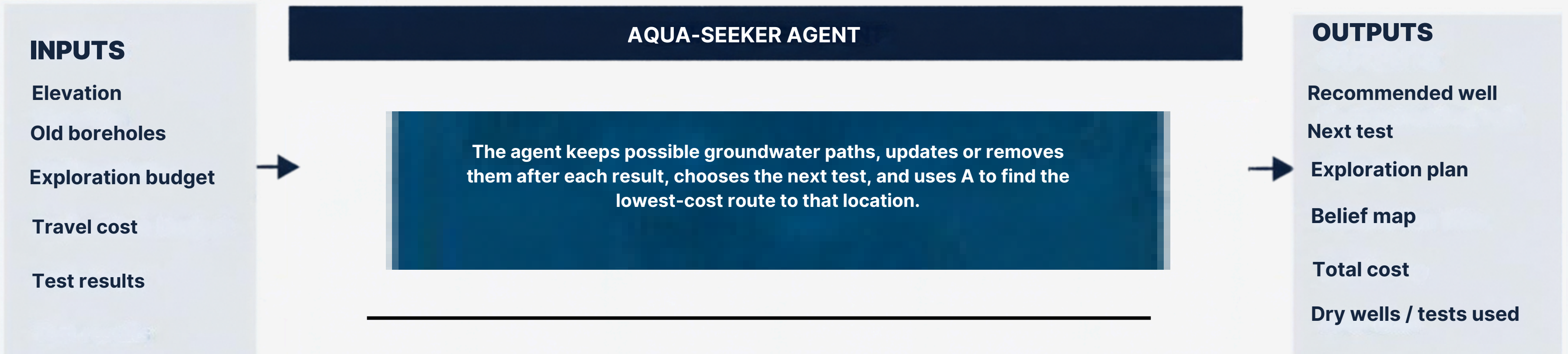


04 Decision
Recommend final site when another test would remove less uncertainty than it costs.



The agent balances **information gain** and **cost** to reach a confident recommendation.

IMPLEMENTATION



AI METHODS USED:

- Intelligent Agents — observes results, updates its knowledge, and chooses the next action.
- A* Informed Search — finds the lowest-cost route to the selected test location.

EVALUATION

We will rigorously test **AQUA-SEEKER** against standard exploration strategies and real-world uncertainty.

OUR RESEARCH QUESTION

Can active test selection reduce unnecessary dry wells compared with static groundwater-potential mapping?

OUR HYPOTHESIS

AQUA-SEEKER becomes more valuable as evidence gets noisier or more incomplete.

WE COMPARE AGAINST

random drilling

vs

fixed-grid testing

vs

map-then-drill

vs

most-likely cell

WE MEASURE WHAT MATTERS

dry wells

total cost

success within budget

tests used

AQUA-SEEKER is not trying to replace groundwater science. It acts on top of it, turning available evidence into a **sequence of smarter exploration decisions**.

VISION 2030 GOAL: ENSURE SUSTAINABLE USE OF WATER RESOURCES

AQUA-SEEKER APPROACH

REDUCING UNNECESSARY DRY WELLS	Avoids wasting expensive drilling resources
SUPPORTING BETTER GROUNDWATER DECISIONS	Helps identify where there is enough evidence to drill
USING AI FOR SUSTAINABILITY	Applies intelligent decision making to real life Saudi challenges

